Docking bay Specification

Document number	SHV-HB-SPEC-0102
Revision	C
Status	APPROVED
Classification	INTERNAL
Author	Quell Barrowe
Approved by	Dorian Hask
Date	2022-11-06

1. Purpose

This specification defines the performance, interface and verification requirements for the docking bay (HB-DBY) of the Docking & Hangar Bays on Starhaven Orbital Habitat & Shipyard.

2. Scope

Fabrication and installation are performed under work package WP-SHV-HB-03 (WBS 1.6.3), owned by Daro Ardent.

The prime supplier for flight hardware is Cygnus Spaceworks. Responsible engineer: Quell Barrowe.

3. Applicable documents

Document	Title
SHV-SYS-SPEC-0001	Starhaven Master System Specification
SHV-HB-DWG-0202	Docking bay general arrangement drawing
SHV-HB-BOM-0302	Docking bay bill of materials
SHV-QA-PLN-0004	Starhaven Verification and Test Plan
SHV-QA-STD-0002	Station Workmanship Standard
SHV-ICD-PD-HB	Interface Control Document: Power Distribution / Docking & Hangar Bays
SHV-ICD-HB-SY	Interface Control Document: Docking & Hangar Bays / Shipyard Slipways
SHV-ICD-HB-CN	Interface Control Document: Docking & Hangar Bays / Computer Core & Data Network
SHV-ICD-SF-HB	Interface Control Document: Superstructure & Frame / Docking & Hangar Bays

4. Description

The docking and hangar bays receive, service and launch passenger shuttles, cargo haulers and visiting ships behind magnetic atmosphere containment fields.

The docking bay is delivered as a set of line-replaceable assemblies, integrated on station by Construction Operations and verified by Quality Assurance & Test.

5. Performance requirements

ID	Requirement	Value	Tolerance	Verification
HB-DBY-R01	The docking bay shall provide a clear span of 237 m.	237 m	±1%	Analysis

ID	Requirement	Value	Tolerance	Verification
HB-DBY-R02	The docking bay shall provide a mooring load rating of 1,595 t.	1,595 t	±5%	Test

5.1 General requirements

ID	Requirement	Verification
HB-DBY-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.	Inspection
HB-DBY-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.	Analysis
HB-DBY-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.	Inspection
HB-DBY-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.	Analysis
HB-DBY-G05	Documentation: Acceptance data packages shall be delivered with every unit.	Inspection

6. Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

7. Verification

Each requirement is verified by the method stated in section 5, following SHV-QA-PLN-0004. Acceptance test reports are filed under the HB test-report series.

8. Revision history

Rev	Date	Description	Author
A	2021-12-15	Initial release.	Quell Barrowe
B	2022-05-13	Incorporated interface changes.	Quell Barrowe
C	2022-11-06	Clarified verification methods.	Quell Barrowe